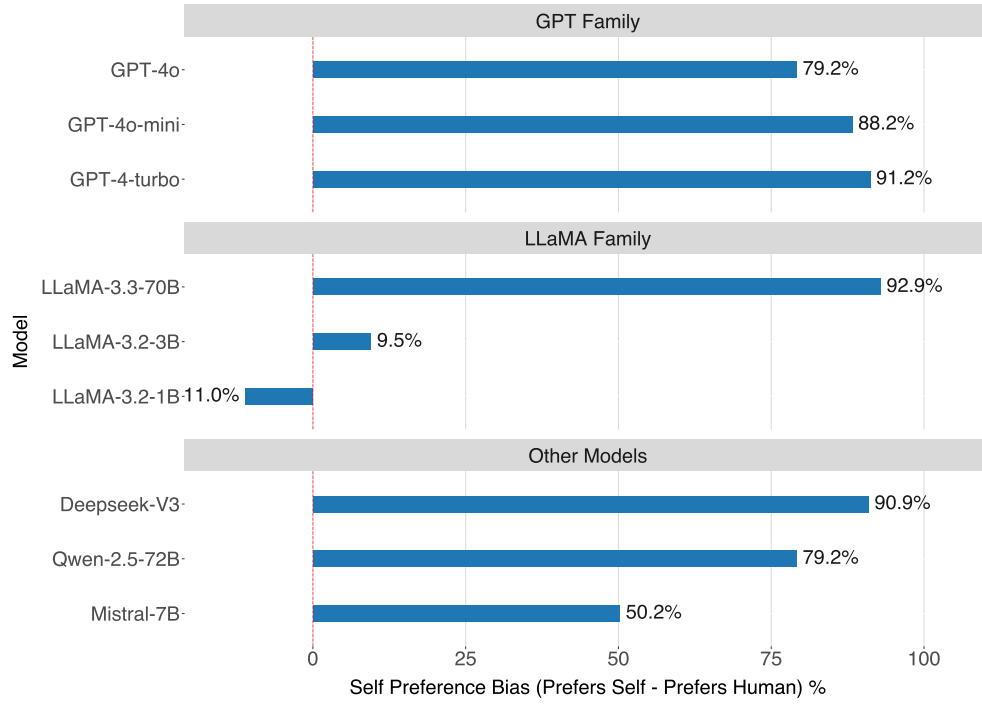
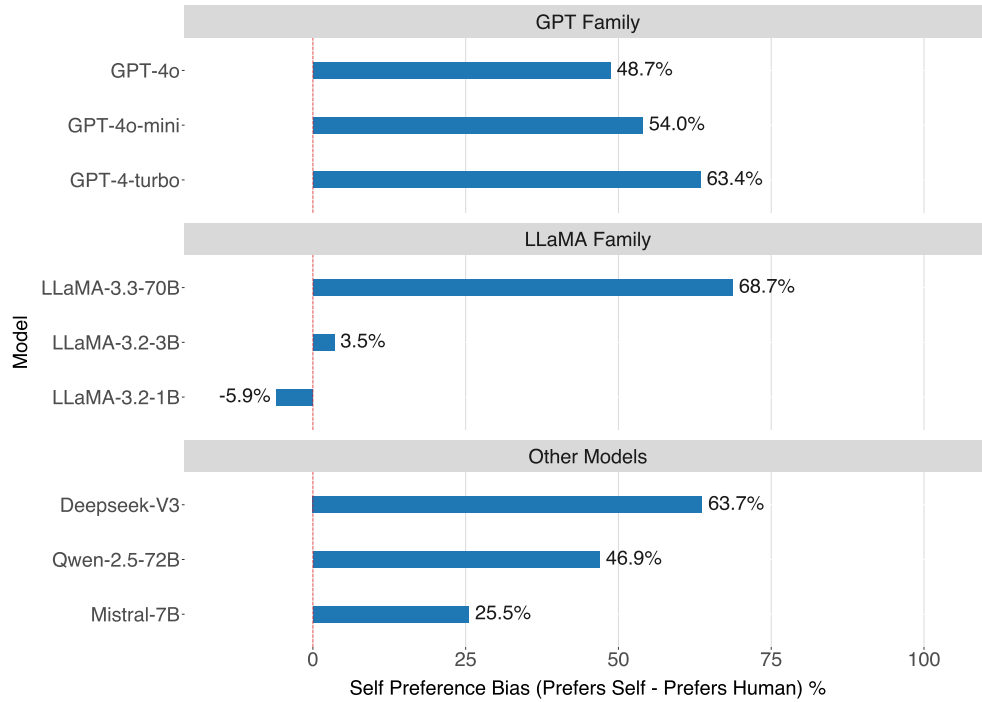




(a) LLM-vs-Human Statistical Parity Self-preference Bias



(b) LLM-vs-Human Equal Opportunity Self-preference Bias

Figure 5 LLM-vs-Human Self-Preference Bias under Resume Revision Condition

approximately 23.2% relative to GPT-4o and 8.5% relative to LLaMA-3.3-70B—magnitudes that are comparable to, and in some cases larger than those observed under resume summary reconstruc-

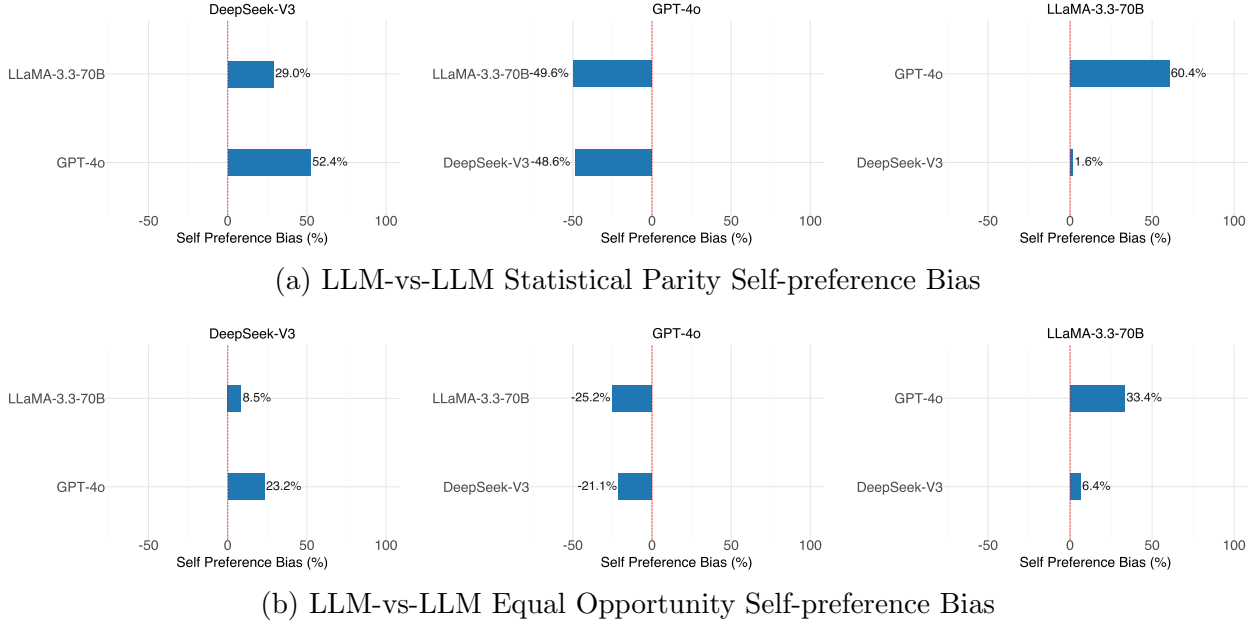


Figure 6 LLM-vs-LLM Self-Preference Bias under Resume Revision Condition

tion. In addition, LLaMA-3.3-70B also exhibits substantial self-preference when evaluated against GPT-4o, indicating that asymmetric self-preference across model pairings persists even under the revision-based counterfactual.

5.4. Impact of Self-Preference Bias in Algorithmic Hiring

To quantify the practical implications of self-preference bias, we simulate a resume-screening pipeline modeled on competitive job markets. The simulation covers 24 occupational categories, each with 30 runs. In each run, we sample five candidate profiles and construct paired resumes: one human-written summary and one counterfactual summary generated by the evaluator LLM itself. These ten resumes (five human, five AI) form a candidate pool competing for four interview slots. The pool is randomly shuffled and presented to an evaluator model (GPT-4o, DeepSeek-V3, or LLaMA-3.3-70B), which is required to return exactly four finalists in ranked order. Since the AI-generated summaries are counterfactuals of the human-written ones, substantive content is held constant and we effectively control for resume quality. Thus, in the absence of bias, human and AI resumes should be equally likely to be selected—on average, two of each from every pool of ten candidates. For each run, we record the number of human versus AI resumes shortlisted, and then aggregate results across runs within each job category. This allows us to compute the magnitude of self-preference bias and construct 95% confidence intervals at the category level.

The results shown in Figure 7 reveal a systematic bias in favor of candidates who use the evaluator LLM to craft their resumes. In the absence of bias, we would expect resumes to be selected evenly, two human-written and two evaluator LLM-generated out of every four interview slots, so the

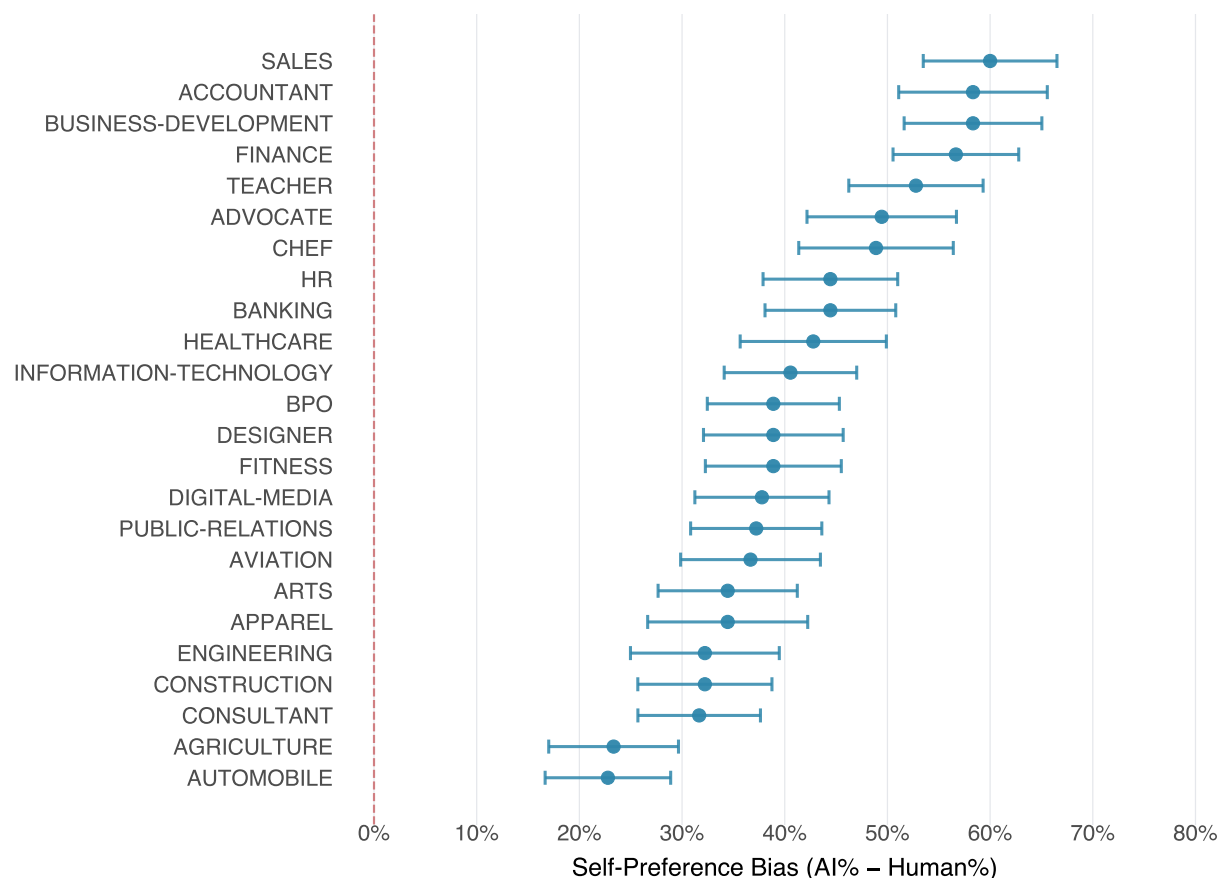


Figure 7 Self-Preference by Job Category

Notes: Each bar shows the average self-preference bias across three evaluator LLMs (GPT-4o, DeepSeek-V3, and LLaMA-3.3-70B) in simulated hiring pipelines. Results are averaged across models; outcomes for individual models are reported in the Appendix H. Positive values indicate that candidates using the evaluator LLM are more likely to be shortlisted than those submitting human-written resumes. Across occupations, evaluator-generated resumes are consistently overrepresented among those selected.

probabilities in the figure would cluster around zero. Positive values indicate that evaluator LLM-generated resumes are more likely to be shortlisted, while values below zero would indicate the opposite. According to our results, however, all values lie well above zero, showing that evaluator-generated resumes are consistently overrepresented among those selected. On average across the three evaluator models, candidates using the evaluator LLM are 23% to 60% more likely to be shortlisted than those submitting human-written resumes. The disadvantage for human-written resumes is most pronounced in business-related occupations such as sales and accountant, and least evident in fields such as automobile and agriculture. Repeating the pipeline 30 times per job category yields consistent results, with confidence intervals that exclude zero in every case.

The consequences extend beyond individual selection outcomes. If access to certain LLMs is uneven across socioeconomic or linguistic groups, this bias risks amplifying existing inequities in